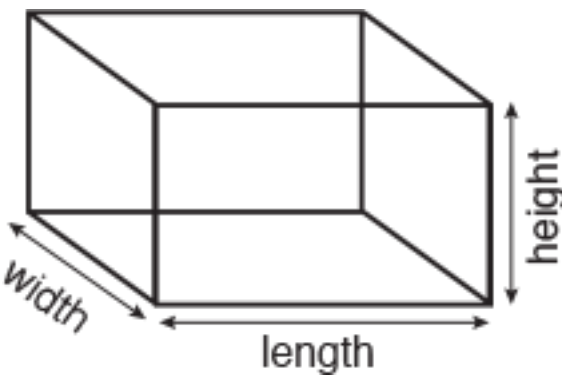


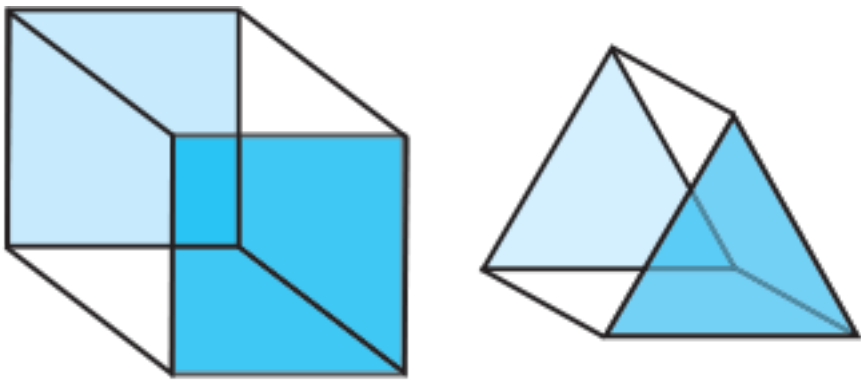
# Prisms

A line has only one dimension—length. A flat shape has two dimensions—length and width. Most objects in life have three dimensions—length, width, and height. Depending on the situation, one of these dimensions may be called *depth*.



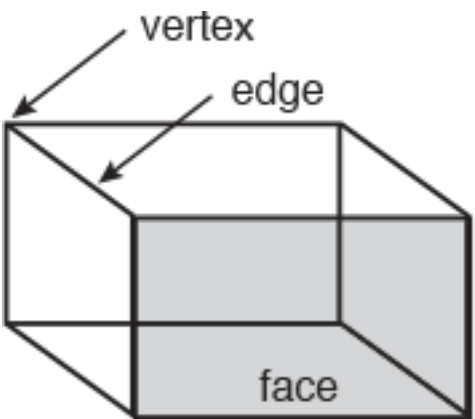
A **prism** is a three-dimensional object with parallel, identical bases that have straight edges. The sides, or faces, that connect the bases are always rectangles.

The drawing shows a rectangular prism and a triangular prism. Notice that the base determines the type of prism.



A corner of the prism is a **vertex**. The **edges** are the straight segments that define the prism’s shape. The flat side of a prism is also called its **face**.

In the drawing, the prism has 8 vertices, 12 edges, and 6 faces. Take a moment to find them all.



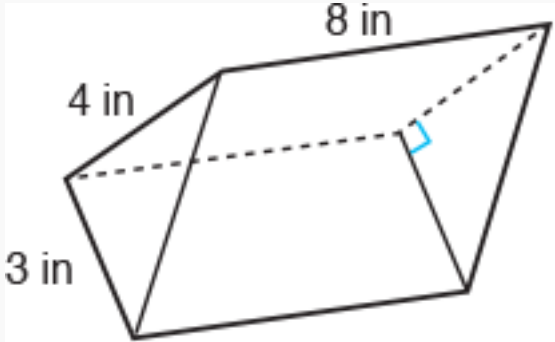
**Volume** is the measure of the space inside a three-dimensional object. Volume is measured using cubic units. To find volume, you need to figure out how many cubes will fit inside the figure. Use the formula below to find the volume of any prism.

**Volume (*V*) of a Prism**     $V = Bh$ , where *B* is the area of one base and *h* is the height

## Example

[Hide](#)

Find the volume of the prism.



### Step 1

Identify the bases.

Remember, bases are identical and parallel. The bases are right triangles.

### Step 2

Find the area of one base.

$$\begin{aligned} A &= \frac{1}{2}bh \\ &= \frac{1}{2} \cdot 4 \cdot 3 \\ &= 6 \text{ sq in} \end{aligned}$$

### Step 3

Multiply by the height, 8 inches.

$$6 \cdot 8 = \mathbf{48 \text{ cubic inches}} \text{ or } \mathbf{48 \text{ in}^3}$$

## Key Terms

**edge**

**face**

**prism**

**volume**

## Rate Your Confidence

High

Medium

Low

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## Category

Lesson 4: Volume

**Related Questions:**  
[Practice questions taken related to this topic](#)